Day 65 — Final Enhancements & Feature Polish

Changes in this batch

• Added `utils/recruiter_report_formatter.py`, a presentation-only adapter that validates required fields and produces a consistent recruiter-facing report.

• Added `utils/ai_observability.py` with structured JSON event records, sensitive-key redaction, in-process counters/latency snapshots, configurable alert threshold evaluation and metadata-only decision audit records.

• Added tests for both helpers and a synthetic end-to-end demo dataset.

Scoring consistency

No existing production scoring weights or decision thresholds were changed. The existing scoring engines remain authoritative. The new formatter validates 0–100 scores and preserves the underlying values; it does not recalculate a recommendation.

Error handling and usability

New helpers reject malformed event metadata, invalid metric values, missing candidate/round fields and invalid scores with clear `ValueError`/`TypeError` messages. Existing legacy APIs and scoring modules were not broadly refactored in this batch to avoid unverified behavioral changes.

Recruiter output

The report adapter includes candidate reference, role, five round scores, aggregate fit score, coverage, recommendation explanation, report version and human-review limitations. It intentionally omits resume text and interview answers.

Production readiness statement

These changes improve consistency and observability foundations, but do not make the system production-ready. Durable logging, external metrics, real alert delivery, persistent audit storage, authentication, encryption, database/queue integration, load testing and legal/privacy review remain deployment requirements.